

Simulated annealing floorplan search

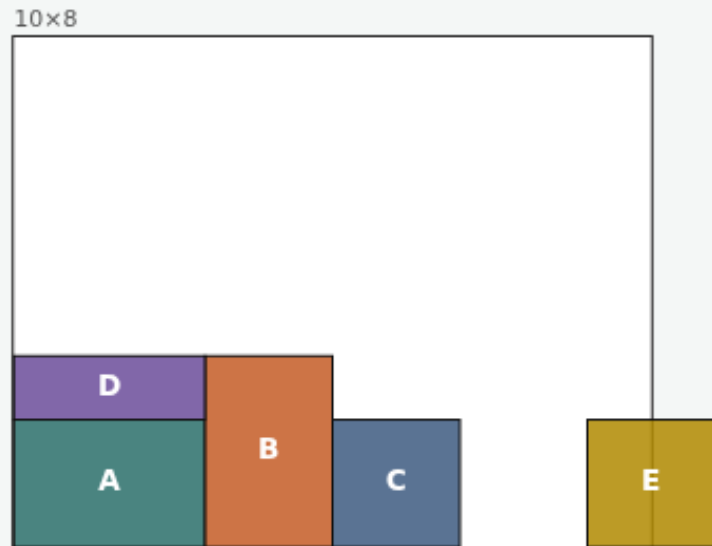
Toy cost adds one thousand when a packing is illegal

Illegal packs pay 1000

SIMULATED ANNEALING FP

Illegal packs pay 1000

Step 1 / 5 · bad-cost · module03-01-simulated-annealing-fp



Explanation

Our toy cost adds one thousand when the packing is illegal. The bad overflow seed therefore sits at cost about one thousand forty-four—dominated by the penalty.

- illegal $\rightarrow +1000$
- Plus deadspace and HPWL terms
- Never accept illegal as “good”

```
cost(bad)≈1044.4  
legal: false
```

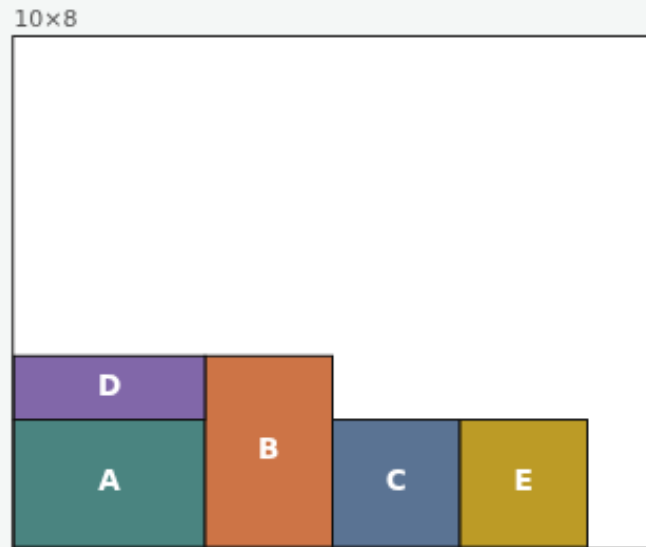
Illegal packs pay 1000

Golden cost stays under 1000

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Golden cost stays under 1000

Step 2 / 5 · golden-cost · module03-01-simulated-annealing-fp



Explanation

The golden legal packing drops below the penalty floor. Cost is about thirty-six—deadspace and a small HPWL proxy, no illegality tax.

- `cost(golden)≈35.7`
- `legal: true`
- Beats bad by ~1000

`cost(golden)≈35.7`
`hpwl≈12.5`

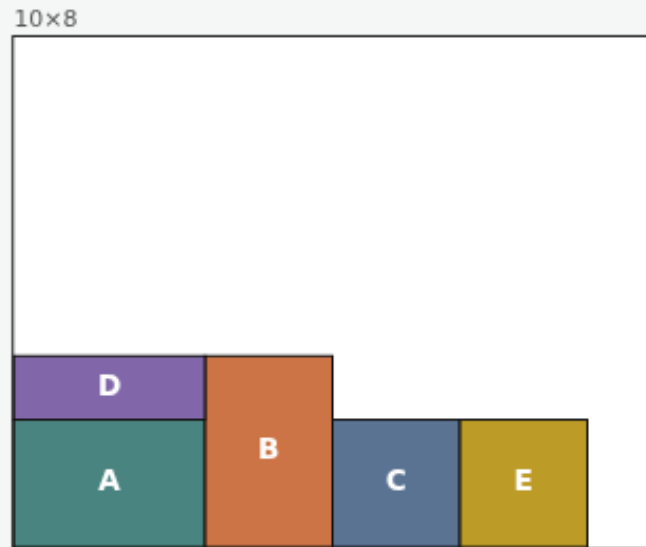
Golden cost stays under 1000

Neighbors swap module positions

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Neighbors swap module positions

Step 3 / 5 · neighbor · module03-01-simulated-annealing-fp



Explanation

A simple SA move swaps the lower-left corners of two modules while keeping sizes. Accept improving moves; accept worsening ones with temperature probability.

- `saSwap(A,E)` exchanges coordinates
- Sizes unchanged
- Re-check legality after the move

move: coordinate swap
then rescore cost

Neighbors swap module positions

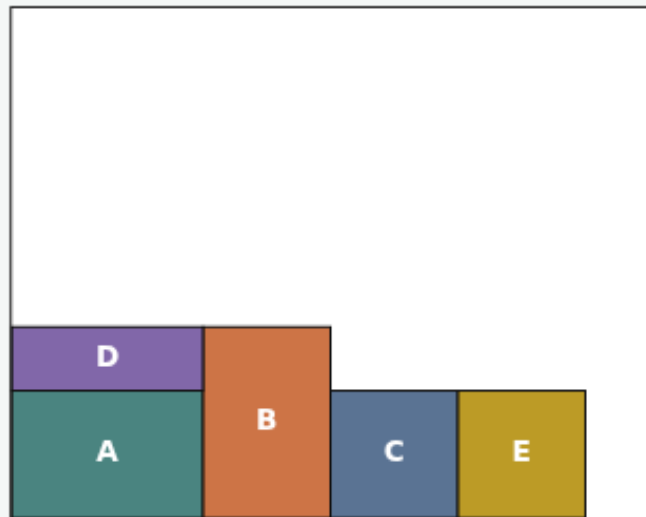
Improve: bad → golden

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Improve: bad → golden

Step 4 / 5 · improve · module03-01-simulated-annealing-fp

10×8



Explanation

One teaching “improve” step replaces the illegal seed with the golden packing. Cost falls below one thousand and legality flips to true—exactly what a cooling schedule should prefer.

- cost drops below 1000
- legal becomes true
- HPWL becomes finite and meaningful

before ≈ 1044.4
after ≈ 35.7

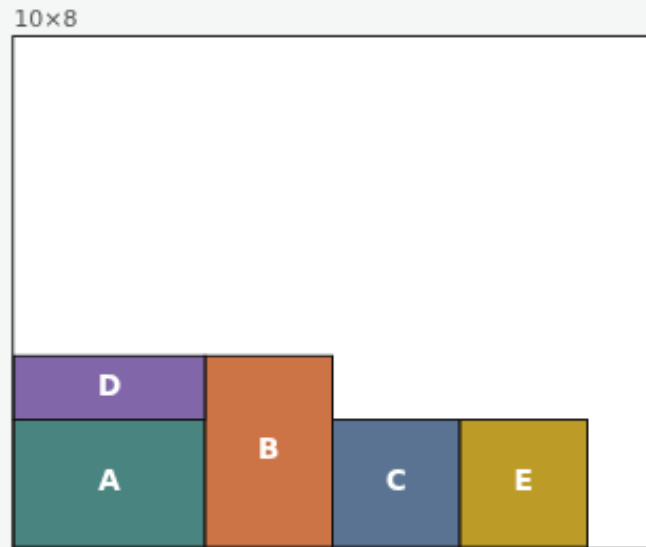
Improve: bad → golden

SA needs a representation

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SA needs a representation

Step 5 / 5 · takeaway · module03-01-simulated-annealing-fp



Explanation

Annealing is only as good as the move set. Pair it with polish, B-star, or sequence-pair edits—plus soft sizing and macros—in the labs that follow.

- Penalty for illegal solutions
- Golden cost $\ll$ bad cost
- Representation defines neighbors

starter: improve to golden

SA needs a representation

Browser lab track

- Open simulated-annealing-fp
- Show bad, note cost at least one thousand
- Show golden or Improve, cost drops below one thousand and legality becomes true

Implement track

- Implement cost with an illegality penalty, plus deadspace and HPWL terms
- Assert $\text{cost}(\text{golden})$ is less than $\text{cost}(\text{bad})$, and saSwap only exchanges coordinates

Pitfalls

- Accepting illegal states without penalty

Your turn

- Demonstrate one improve step from bad to golden
- Next: soft module A reshaped from three by two to two by three